

6(a)(i)	same shape <u>and</u> shift to the right ; <i>must start to rise within 2 small squares of 0 and end at or near to existing curve</i>	1
6(a)(ii)	<i>max 2 if concept of more not mentioned in response</i> <i>any three from:</i> 1 actively respiring tissue means more carbon dioxide ; 2 increase in, formation of carbonic acid / dissociation of carbonic acid / hydrogen ions, (in the red blood cell) ; A from equation 3 more hydrogen ions, bind to haemoglobin / form haemoglobinic acid ; A HHb 4 (causes) more oxygen (to be), unloaded / dissociated / AW (from haemoglobin); I <i>ref. to</i> faster / quicker I incorrect <i>ref. to</i> affinity causing unloading, e.g. CO₂ A lower (percentage) saturation of haemoglobin with oxygen A oxygen released, more easily / readily from haemoglobin 5 haemoglobin <u>affinity</u> for oxygen decreases ; 6 more oxygen to meet demand for (aerobic / cellular) <u>respiration</u> ; AW I more oxygen for respiring tissues	3

Question	Answer	Marks
6(b)	CO₂ = (passive / simple) diffusion ; HCO₃⁻ = <u>facilitated</u> diffusion ; I ref. to size CO₂ is, non-polar / not charged / not ionic <u>and</u> can cross, hydrophobic core / phospholipid bilayer ; HCO₃⁻ is, charged / ionic / hydrophilic <u>and</u> (needs to) cross, via, transport / carrier, protein ; A channel protein <i>if explanation mps not gained, allow 1 mark for CO₂ is, non-polar / not charged / not ionic <u>and</u> HCO₃⁻ is, charged / ionic / hydrophilic</i>	4